Electromagnetic Attack

112-BOLCEW03

Enabling Learning Outcome

The Student will define divisions of Electromagnetic Warfare doctrine, in a classroom environment, given a network computer workstation, training aids, and appropriate references by demonstrating understanding of the divisions of Electromagnetic Warfare Doctrine; Electromagnetic attack, Electromagnetic protect, and Electromagnetic support in order to support the application of EW effects critical to the units mission in accordance with JP 3-13.1, ADRP 6-0, FM 3-12, FM 6-0, ATP 3-36, while achieving a minimum of 70% on all Practical Exercises (PE) and end of module exam.

Learning Step Activities

- Define Electromagnetic attack.
- Describe the fundamentals of (EA).
- Compare and Contrast defensive and offensive EA
- Describe the 7 EA missions
- Describe the fundamentals of jamming
- Compare and Contrast Jamming techniques
- Describe various types of EM Countermeasures

Electromagnetic Attack

Electromagnetic attack is a function of Electromagnetic warfare involving the use of electromagnetic energy, directed energy, or anti-radiation weapons to attack personnel, facilities, or equipment with the intent of **degrading, neutralizing, or destroying enemy combat capability** and is considered a form of fires (JP 3-13.1)

- EA includes:
  - Actions taken to **prevent or reduce an enemy's effective** use of the electromagnetic spectrum (EMS)
  - Employment of weapons that use **either electromagnetic or directed energy** as their primary destructive mechanism: Lasers, RF, Particle Beams
  - **Offensive and defensive** activities, including countermeasures

Offensive vs. Defensive EA

Comparison

- Actions related to EA are either offensive or defensive. Though they are similar actions and capabilities, they differ in purpose.
- Defensive EA attacks an enemy capability to protect friendly personnel and equipment or platforms.
- Offensive EA attacks an enemy capability to effect enemy personnel and equipment or platforms.

Offensive EA Uses:

- Jamming Electromagnetic C2.
- Using anti-radiation missiles to suppress enemy air defenses (anti-radiation weapons use radiated energy emitted from a target as the mechanism for guidance onto the target).
- Using directed-energy weapons to deny, disrupt, or destroy equipment or capabilities.

Defensive EA Uses:

- Aircraft protection measures such as the use of expendables (flares and active decoys), jammers, towed decoys, directed-energy infrared countermeasures
- Counter Radio-controlled Improvised Explosive Device Systems

Electromagnetic Attack Actions

EA **denies, degrades, disrupts, destroys or manipulates (D4M)** enemy combat capability through various of actions

- Actions include:
  - Countermeasures
  - Electromagnetic deception
  - Electromagnetic intrusion
  - Electromagnetic jamming
  - Electromagnetic probing
  - Electromagnetic pulse (EMP)

Countermeasures

Countermeasures is a form of military science that, by the employment of devices and/or techniques, has as its objective the impairment of the operational effectiveness of enemy activity.

- Can be active or passive
- Can be deployed preemptively or reactively.

Include:

- Electro-optical-infrared
- Radio Frequency

Electromagnetic Deception

Deception presents the signatures to the threat providing a false environment.

The three types of deception are:

- **Simulative:** attempts to represent friendly notional or actual capabilities to mislead threat forces

- **Manipulative:** uses communication or non-communication signals to convey indicators that mislead the enemy

- **Imitative:** mimics threat emissions with the intent to mislead them

Electromagnetic probing

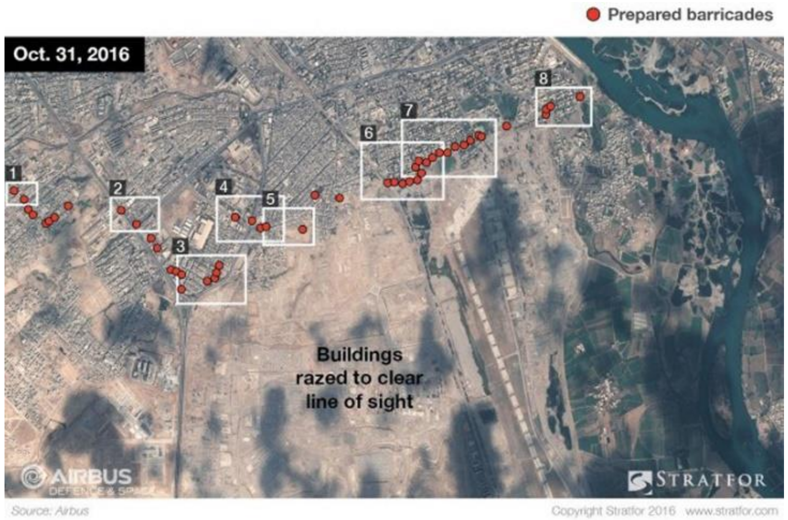
Intentional radiation designed to be introduced into the devices or systems of potential enemies for the purpose of learning the functions and operational capabilities of the devices or systems.

- This activity is coordinated through joint or interagency channels and supported by Army forces.

Electromagnetic Intrusion

The intentional insertion of electromagnetic energy into transmission paths in any manner, with the objective of:

- Deceiving operators
- Causing confusion



Electromagnetic pulse

Electromagnetic radiation from a strong pulse of Electromagnetic energy that when coupled with electrical or Electromagnetic systems produces damaging current and voltage surges.

- An electromagnetic pulse induces high currents and voltages in the target system, damaging electrical equipment or disrupting its function.
- An indirect effect of an electromagnetic pulse can be electrical fires caused by the heating of electrical components.

Electromagnetic Jamming

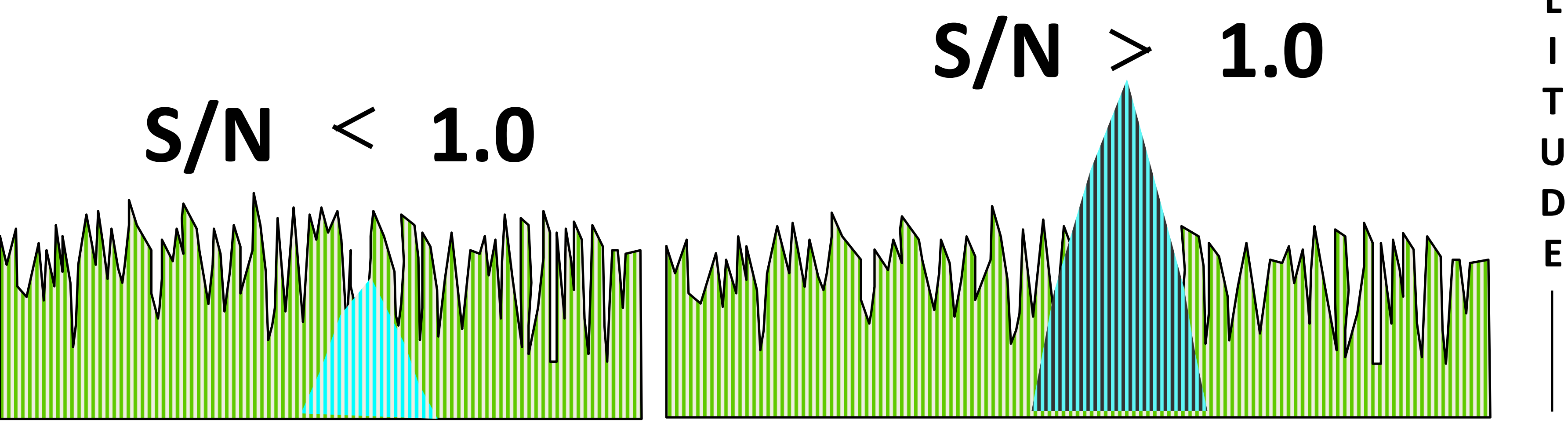
Form of EA that seeks to disrupt the enemy's ability to receive electromagnetic signals by overwhelming the targeted receiver with intentional interference.

- Standoff jamming - supports operations by disrupting or degrading threat command and control systems and sensors
- Escort jamming - is defensive in nature and protects maneuver forces from threat weapons systems that use radio frequency receiving triggers
- Spot jamming - jamming one specific frequency
- Sweep jamming – cover a frequency range but not the specific frequency in use
- Barrage jamming - frequency range but not the specific frequency in use

Electromagnetic Jamming

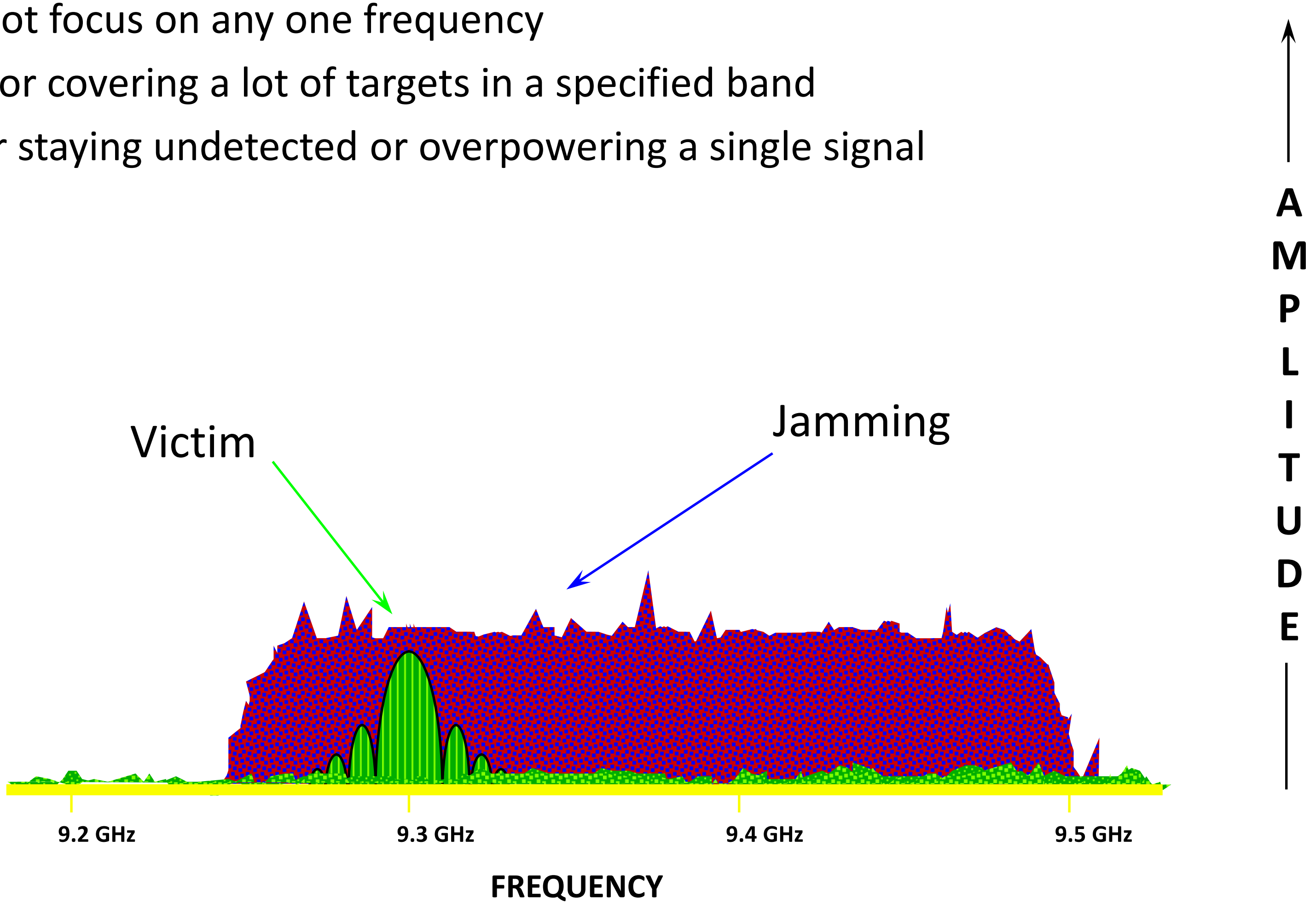
Signal to Noise Ratio (S/N)

- Ratio of a signal compared to local noise.
- For any receiver to accurately detect a signal, the S/N ratio must be greater than one.
- If S/N ratio is less than one, the receiver will not be able to detect the signal (jammed)
  - **The basic principle of jamming is to create a signal that raises the level of the noise at a receiver, so it is greater than that of the incoming radio signals.**
  - **Jamming affects receivers not transmitters!**



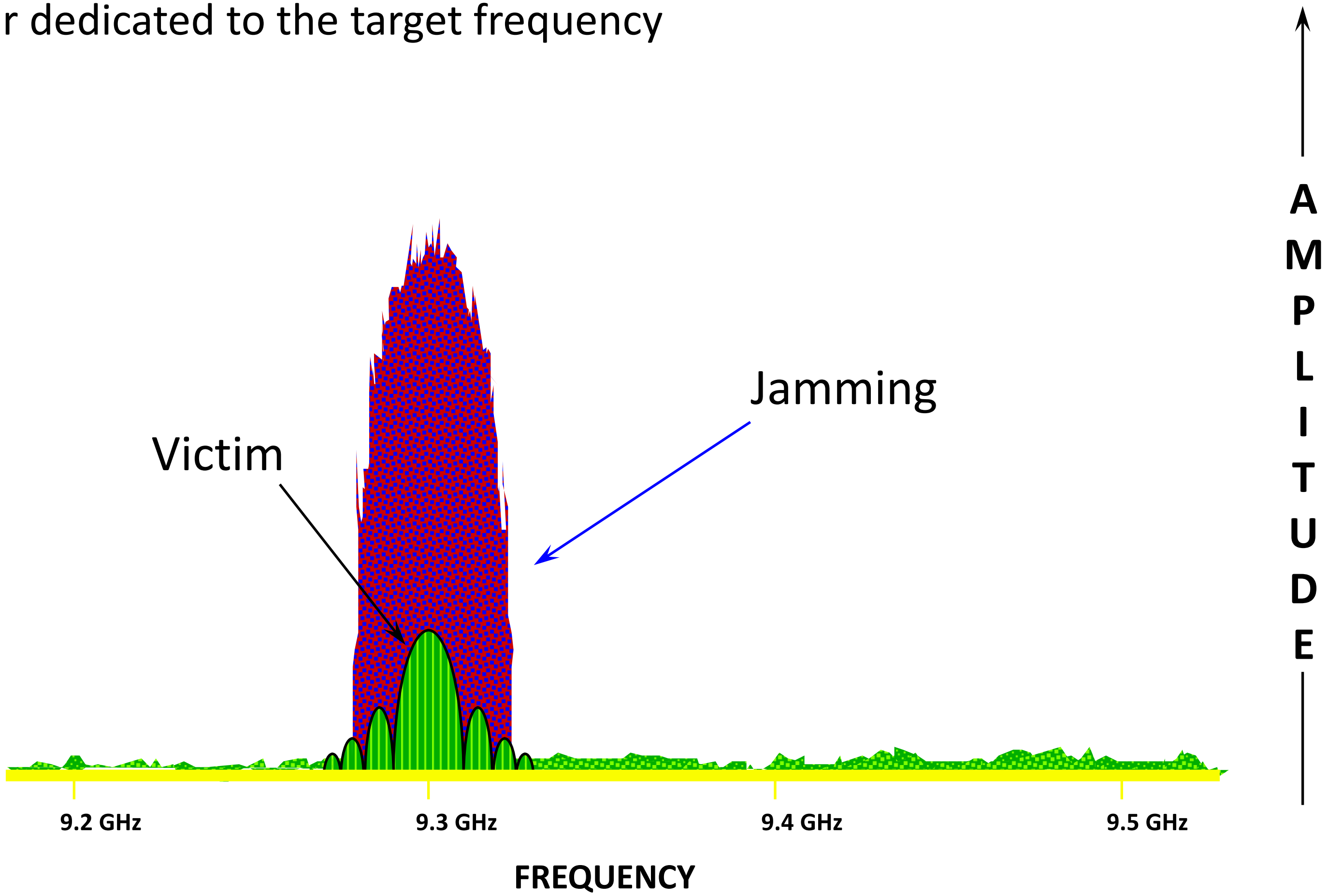
Barrage Noise Jamming

- Covers large bandwidth
- Disperses power over entire bandwidth (requires a lot of power to be effective)
- Does not focus on any one frequency
- Good for covering a lot of targets in a specified band
- Bad for staying undetected or overpowering a single signal



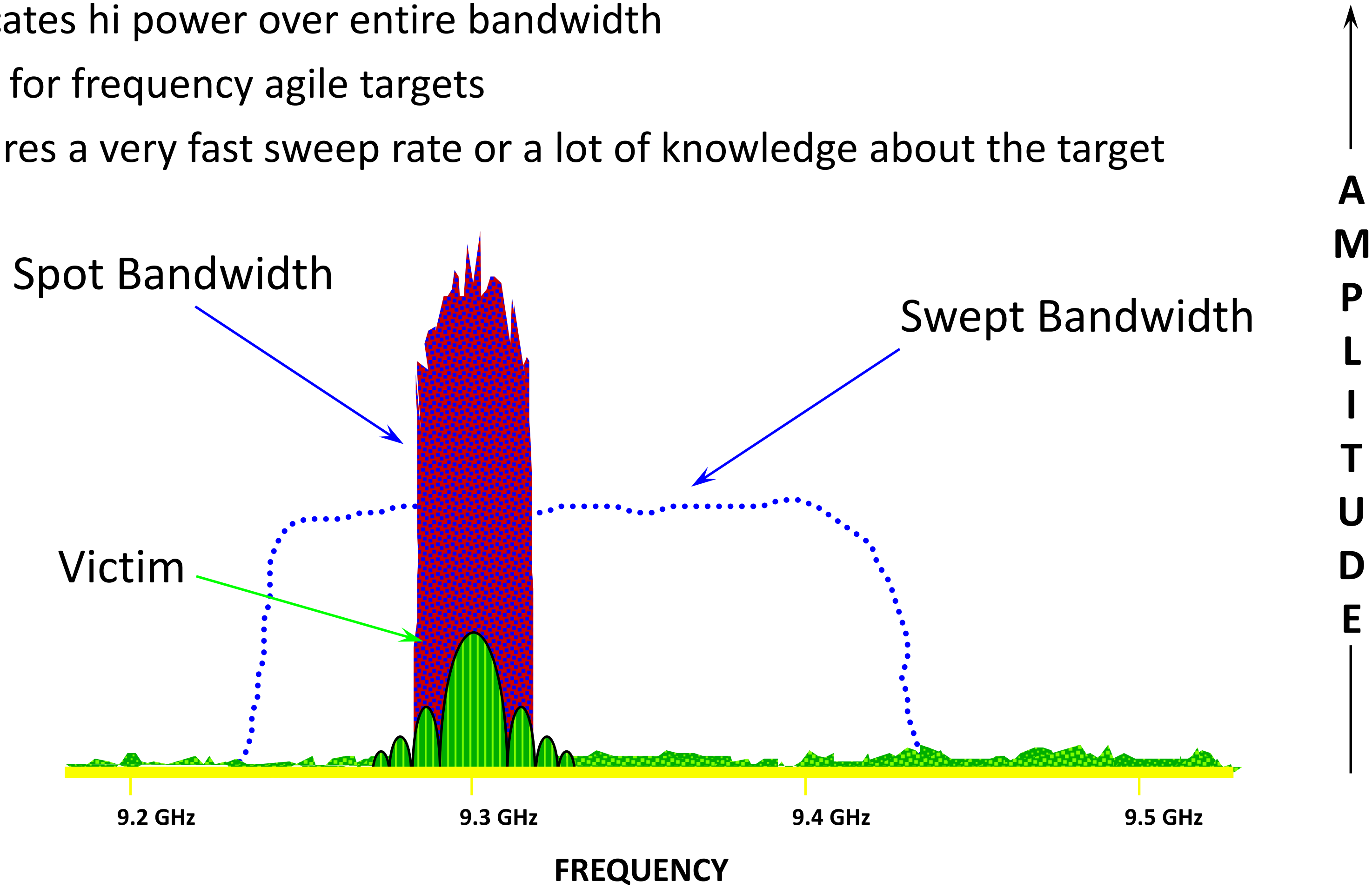
Spot Noise Jamming

- Small bandwidth
- Effective against a single known frequency
- Ineffective against frequency agile signals (hopping)
- More power dedicated to the target frequency



Swept Spot Noise Jamming

- Covers large bandwidth by constantly changing frequency
- Does not provide continuous coverage of the entire band
- Dedicates hi power over entire bandwidth
- Good for frequency agile targets
- Requires a very fast sweep rate or a lot of knowledge about the target



Special Case (Radar Jamming)

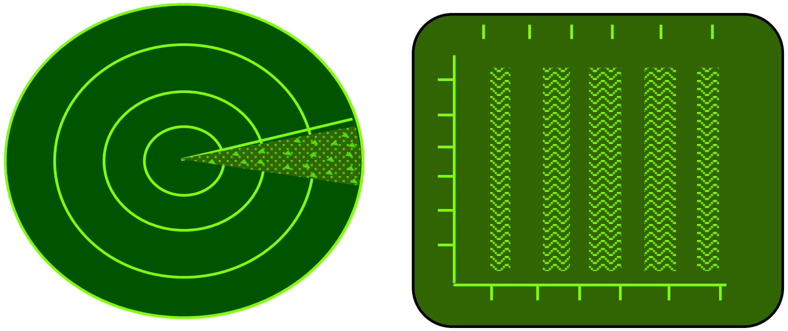
- Active Radar Jamming
  - Takes advantage of radar receiver sensitivity and transmission pattern of the radar antenna to deny critical information to the victim radar.

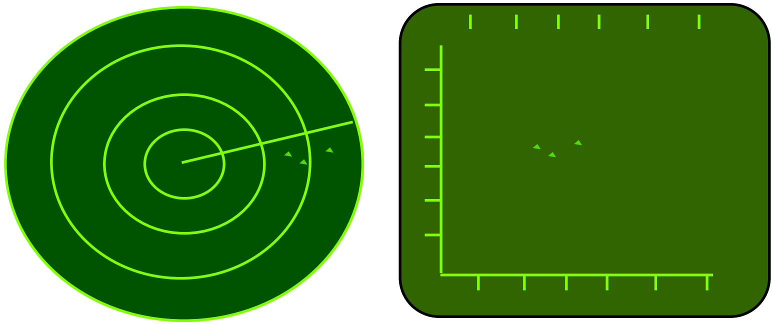
NOISE JAMMING

RF energy transmitted on the victim's frequency intended to interfere with receiver gain

DECEPTION JAMMING

RF energy transmitted to deceive the victim radar discriminators or operator

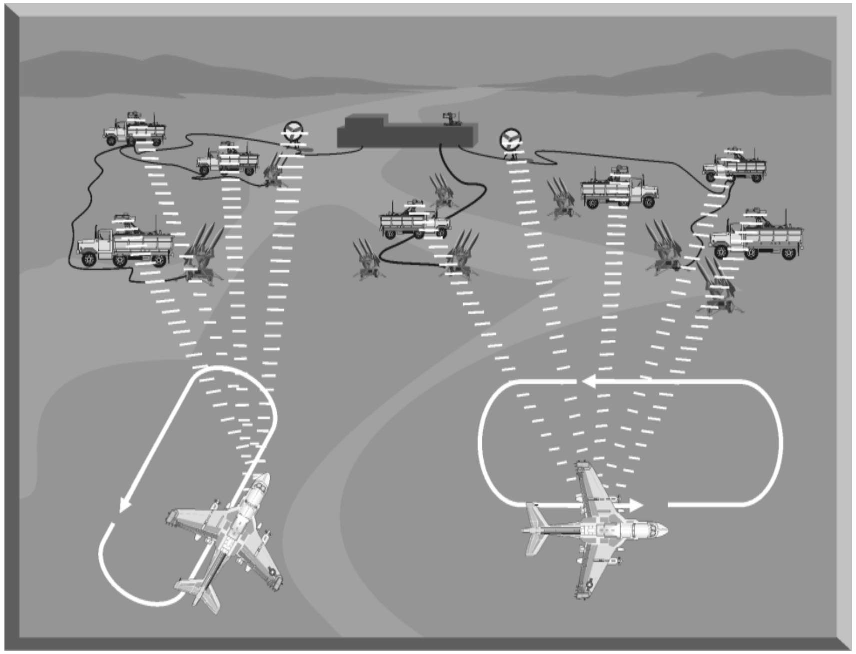


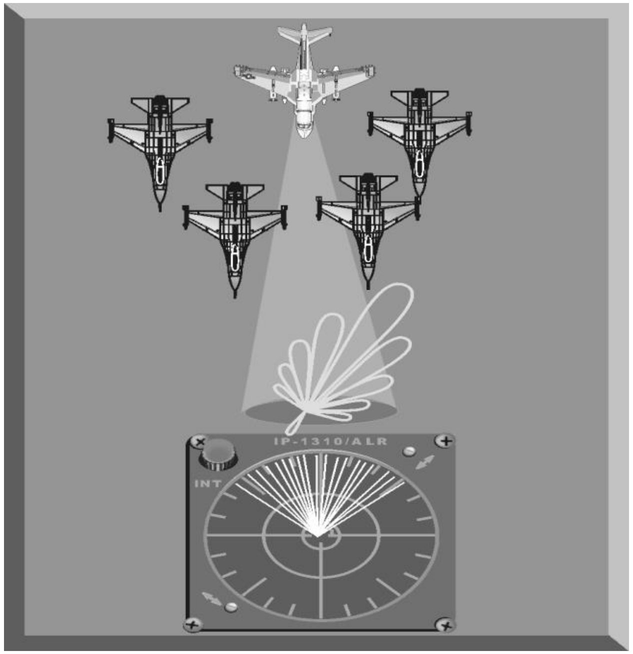


Special Case (Radar Jamming)

Two primary employment options for both noise and deception jamming techniques

1. Support jamming
   - Stand-off jamming (SOJ)
   - Escort jamming
2. Self-protection jamming



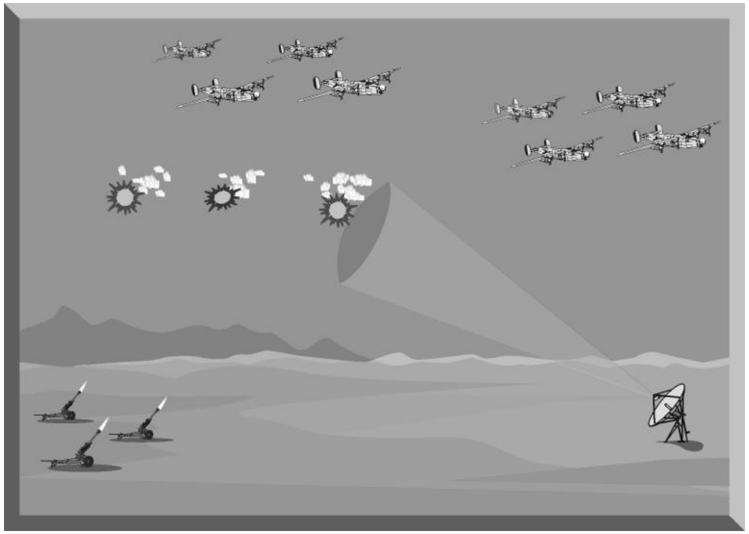


Countermeasures

Chaff

UNCLASSIFIED

- Dispensed metallic strips that reflect EM waves
- Most widely used and effective expendable electronic attack (EA) devices
- Dispensed in atmosphere to deny radar acquisition, generate false targets, and to deny or disrupt radar tracking
- Useful against modern radar systems
- First used during World War II
  - Royal Air Force, under code name “WINDOW,” dropped bales of metallic foil during a night bombing raid in July 1943.



Decoys

- A decoy is a device designed to replicate (present an electronic signature) similar or better than the actual target.
- Decoys do three primary missions:
  - Saturate the enemy's radar/detection system
  - Coerce the enemy into exposing his forces prematurely
  - Defeat enemy tracking systems

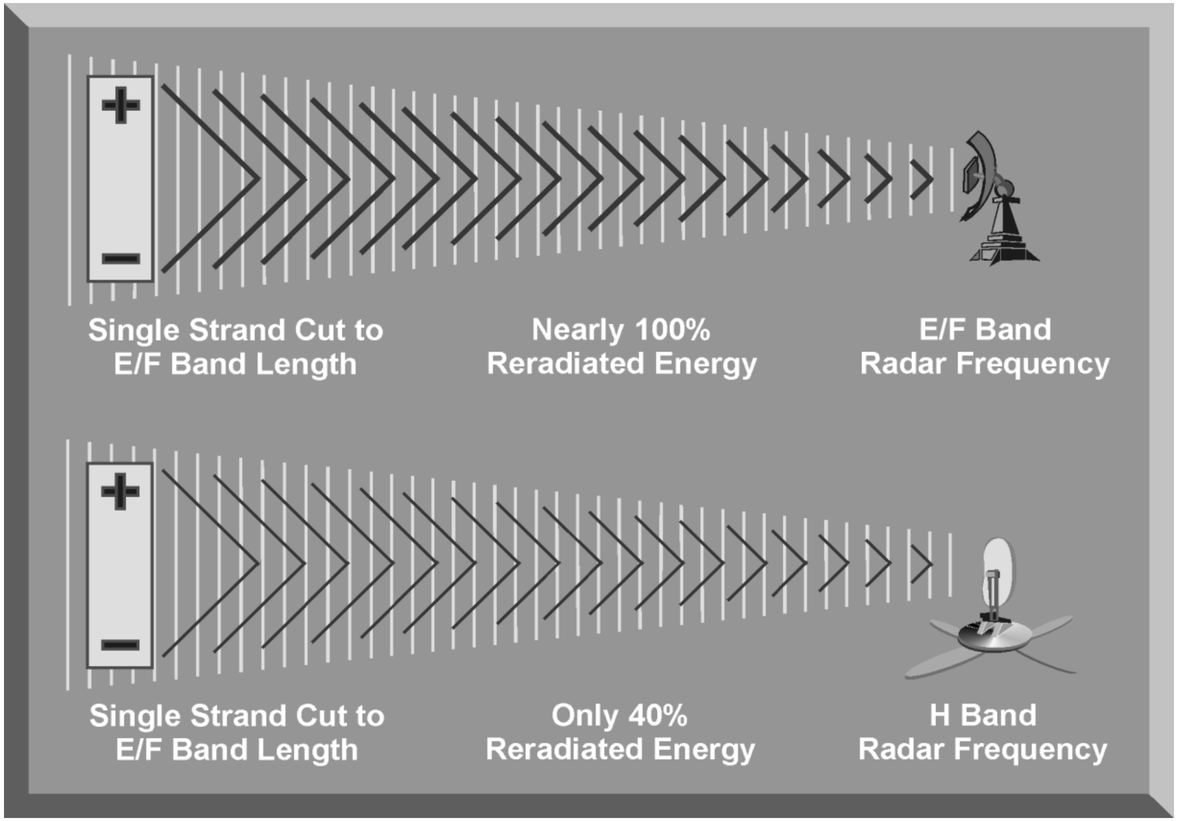
Tactical Air Launched Decoy (TALD)

Miniature Air Launched Decoy (MALD)

Chaff Frequency and Wavelength

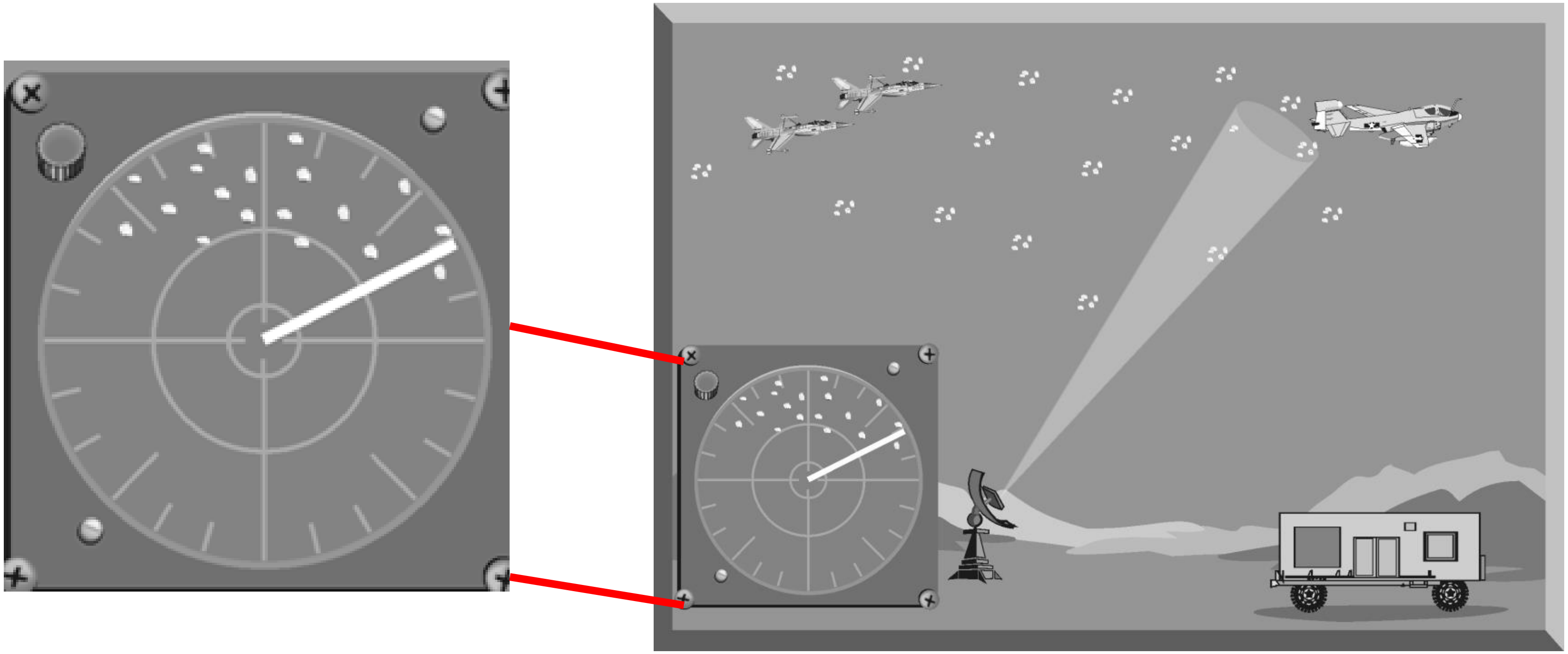
UNCLASSIFIED

- Optimum size
  - Approximately one-half wavelength of victim radar freq
  - Varying lengths packaged together provide coverage over wide frequency range



Area Saturation

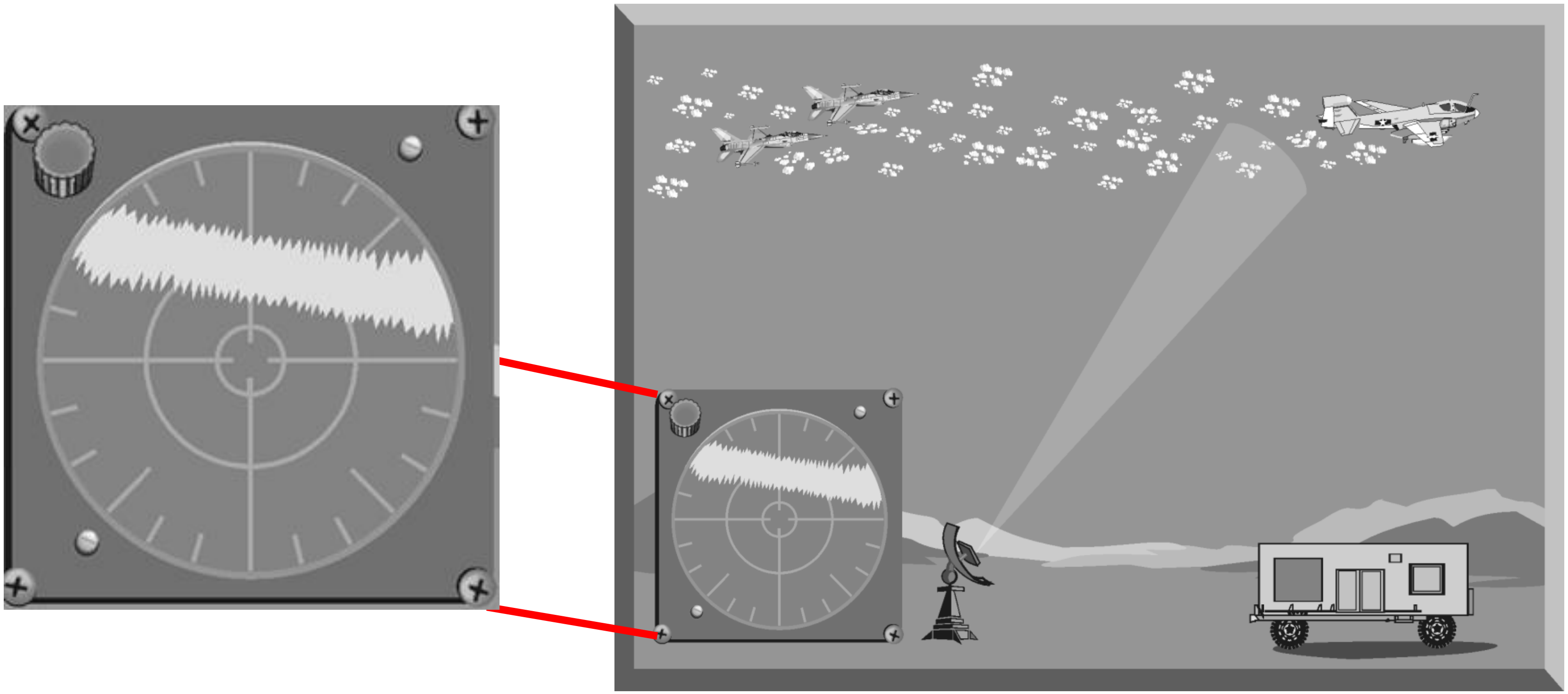
- Present multiple false targets
  - Specific areas
  - Saturate radar systems
  - Confuse the enemy



Chaff Corridor

UNCLASSIFIED

- Screen the ingress and egress of attack package
- Dispensing large chaff quantities in continuous “ribbon”



Electromagnetic Attack Considerations (1 of 2)

Friendly Communications:

- De-confliction with communications and other EMS dependent systems a challenge.

Intelligence Collection:

- EA operations may interrupt intelligence collection by jamming or inadvertently interfering with a particular frequency being used to collect data on the threat

- Signals Intelligence (SIGINT) collectors need to be aware that Cyberspace Electromagnetic Activities (CEMA) elements are charged with controlling the EMS and that this can take precedence over collection

Electromagnetic Attack Considerations (2 of 2)

Other Effects: Information Related Capabilities (IRCs)

- Other effects that rely on EMS when planning for EA: Military Support Operation (MISO) and Military Deception (MILDEC)

Non-hostile Local EMS Use

- EA can affect local media and communications systems and infrastructure

Hostile Intelligence Collection

- A well-equipped enemy can detect friendly EW activities and thus gain intelligence on friendly force intentions.

Persistency of Effect

- Effects of jamming only persist as long as jammer itself is emitting and is in range to affect the target

Summary

- Define Electromagnetic attack.
- Describe the fundamentals of (EA).
- Compare and Contrast defensive and offensive EA
- Describe the 7 EA missions
- Describe the fundamentals of jamming
- Compare and Contrast Jamming techniques
- Describes various types of EM Countermeasures
- Identify Electromagnetic Considerations

Questions?

References

- FM 3-12 Army Cyberspace and Electromagnetic Warfare Operations. Apr 17.
- ATP 3-12.3 Electromagnetic Warfare Techniques.
- JP 3-13.1 Electromagnetic Warfare

Electromagnetic Protection

112-BOLCEW03

Enabling Learning Outcome

The Student will define divisions of Electromagnetic Warfare doctrine, in a classroom environment, given a network computer workstation, training aids, and appropriate references by demonstrating understanding of the divisions of Electromagnetic Warfare Doctrine; Electromagnetic attack, Electromagnetic protect, and Electromagnetic support in order to support the application of EW effects critical to the units mission in accordance with JP 3-13.1, ADRP 6-0, FM 3-12, FM 6-0, ATP 3-36, while achieving a minimum of 70% on all Practical Exercises (PE) and end of module exam.

Learning Step Activities

- Define Electromagnetic Protection.
- Describe the fundamentals of (EP).
- Describe EP tasks
- Describe EP TTPS
- Describe Electromagnetic Interference Reporting
- Describe EP planning considerations

Principles of Electromagnetic Protect (EP)

Electromagnetic Protect (EP) (1 of 3)

Electromagnetic protection is a division of Electromagnetic Warfare involving actions taken to protect personnel, facilities, and equipment from:

- Friendly interference
- Neutral interference
- Enemy interference
- Naturally occurring phenomena

EP Vs Defensive EA

- Both are used to protect friendly units in the EMS
- EP is action taken on friendly systems and capabilities
- Defensive EA is action taken against enemy systems and capabilities

Electromagnetic Protect (EP) (2 of 3)

Electromagnetic protection uses a wide variety of techniques such as:

- Limiting transmissions
- Radio PACE plans
- Natural objects to objects to mask radiated energy from traveling to undesirable destinations (terrain masking)
- Directional antennas
- Spectrum management

Electromagnetic protection is essential to prevent the adversary from learning behavior and intentions within the EMS.

Electromagnetic Protect (EP) (3 of 3)

EP also includes physical security, Communications Security (COMSEC) measures, system technical capabilities,
such as:

- Frequency hopping
- Shielding of Electromagnetics
- EMS management

To establish emission control best practices, the EWO considers the following:

- Vulnerability analysis and assessment of friendly communications assets
- EP monitoring techniques and feedback procedures
- EP effects on friendly capabilities
- Transmit power settings
- Designating user time slots for transmissions

> "The most basic protection from enemy EW is radio discipline. This can take the form of limiting transmissions, antenna masking, and use of low power."
>
> ***Center for Army Lessons Learned***

Electromagnetic Protect (EP) Tasks

- Electromagnetic Hardening
- Electromagnetic Masking
- Emission Control (EMCON)
- Electromagnetic Spectrum Management
- Wartime Reserve Modes (WARM)
- Electromagnetic Compatibility

Electromagnetic Protection Actions (1)

- Electromagnetic Hardening
  - Actions taken to protect personnel, facilities, and equipment by blanking, filtering, attenuating, grounding, bonding, and/or shielding against undesirable effects of electromagnetic energy (JP 3-13.1).
- Electromagnetic masking
  - Controlled radiation of electromagnetic energy on friendly frequencies in a manner to protect the emissions of friendly communications and Electromagnetic systems against enemy ES measures/signals intelligence without significantly degrading the operation of friendly systems (JP 3-13.1).
- Emission Control
  - Selective and controlled use of electromagnetic, acoustic, or other emitters to optimize command and control capabilities while minimizing, for operations security: a) detection by enemy sensors; b) mutual interference among friendly systems; and/or c) enemy interference with the ability to execute a military deception plan (JP 3-13.1).

Electromagnetic Protection Actions (2)

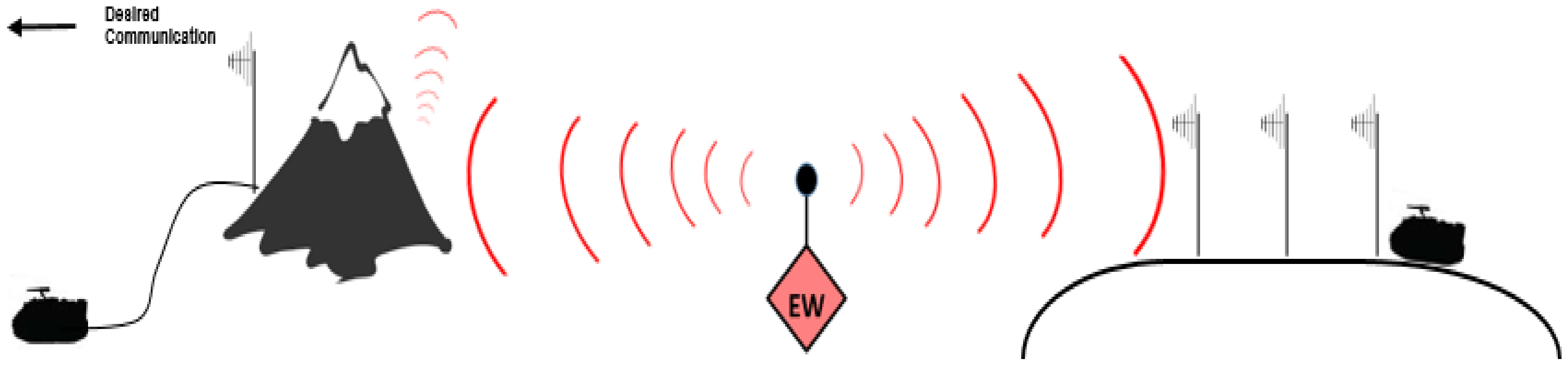
- Electromagnetic Spectrum Management
  - Planning, coordinating, and managing use of the electromagnetic spectrum through operational, engineering, and administrative procedures (JP 6-01).
  - Objective of spectrum management is to enable Electromagnetic systems to perform their functions in the intended environment without causing or suffering unacceptable interference.
- Wartime reserve modes (WARM)
  - Characteristics and operating procedures of sensors, communications, navigation aids, threat recognition, weapons, and countermeasures systems that will contribute to military effectiveness if unknown to or misunderstood by opposing commanders before they are used but could be exploited or neutralized if known in advance (JP 3-13.1).
  - Wartime reserve modes are deliberately held in reserve for wartime or emergency use and seldom, if ever, applied or intercepted prior to such use.

Electromagnetic Protection Actions (3)

- Electromagnetic Compatibility
  - Ability of systems, equipment, and devices that use the electromagnetic spectrum to operate in their intended environments without causing or suffering unacceptable or unintentional degradation because of electromagnetic radiation or response (JP 3-13.1).
  - Electromagnetic Compatibility involves the application of :
    - Sound electromagnetic spectrum management
    - System, equipment, and device design configuration that ensures interference-free operation
    - Clear concepts and doctrines that maximize operational effectiveness

EP TTPs

Electromagnetic Protection TTPs (Terrain Masking)

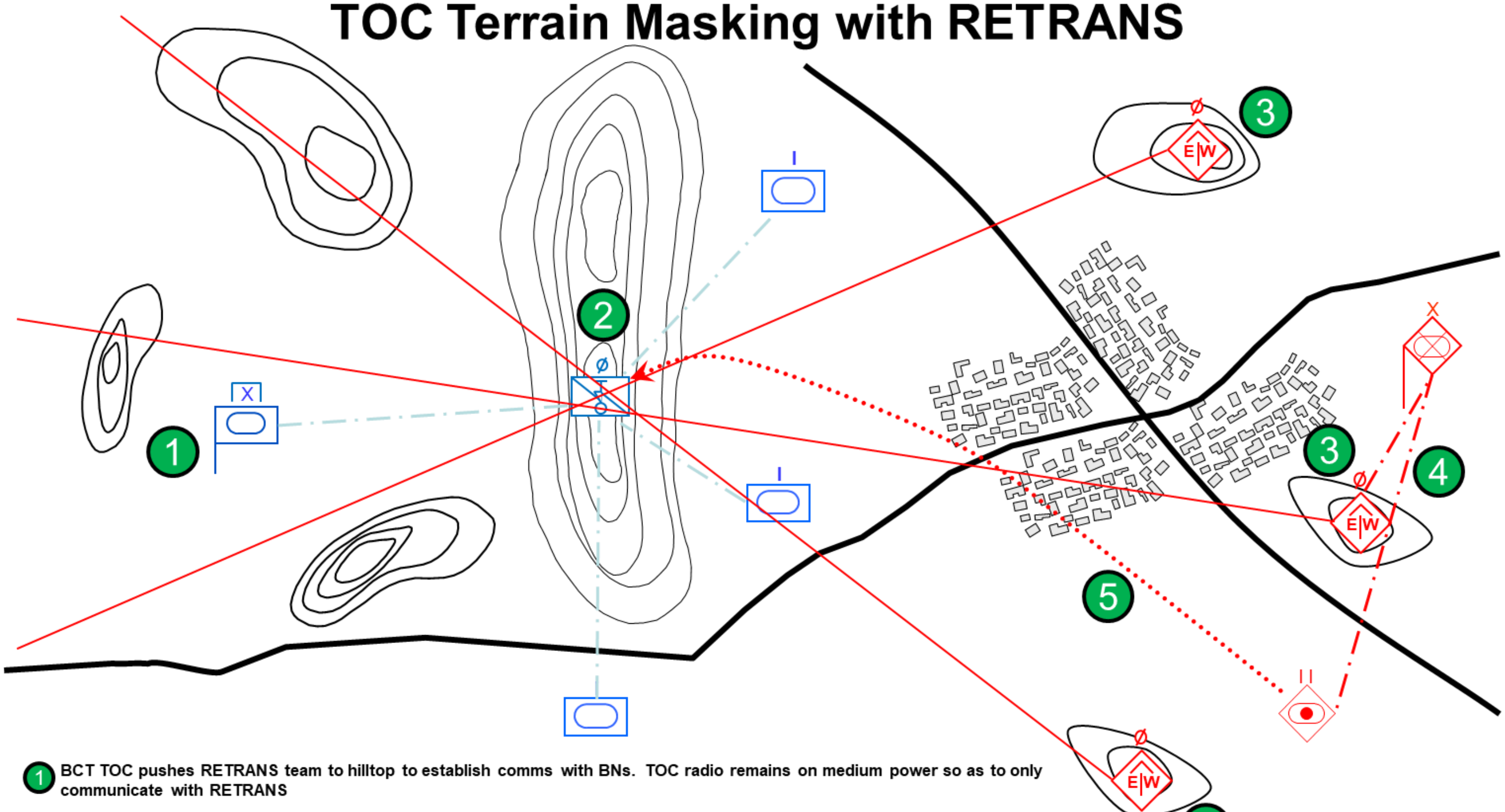


Which CP Would You Rather Be In

- What EP measures are shown?
- How can you integrate these principals into traditional TOC set-ups or layouts?

Electromagnetic Protection TTPs (Terrain Masking)

TOC Terrain Masking with RETRANS

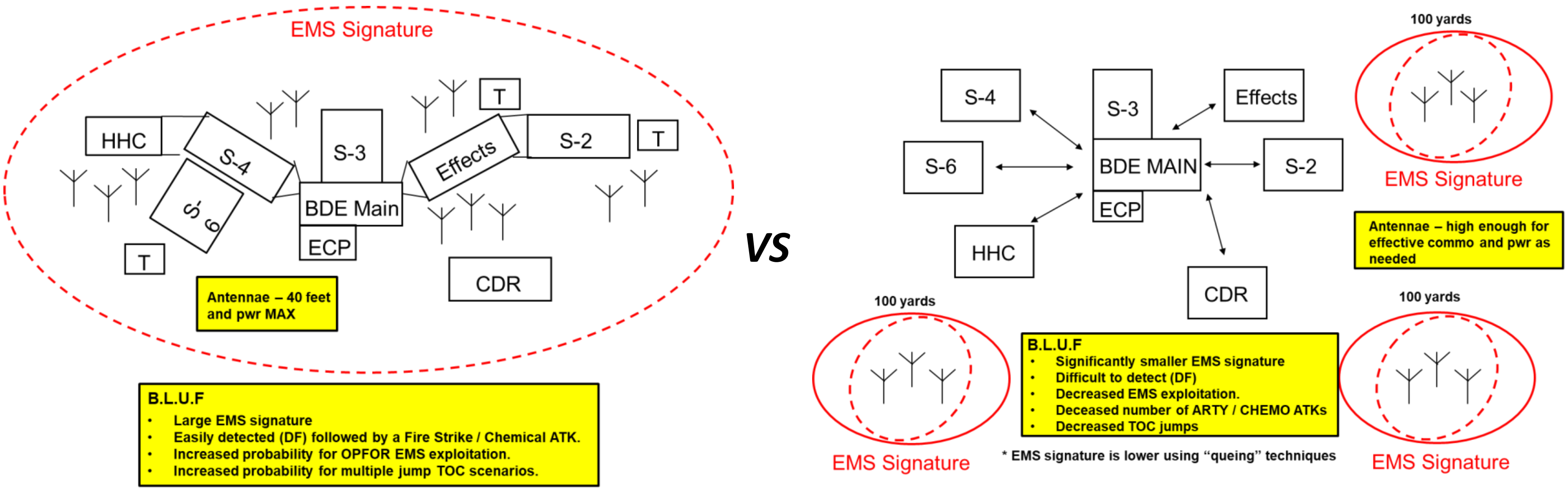


1. BCT TOC pushes RETRANS team to hilltop to establish comms with BNs. TOC radio remains on medium power so as to only communicate with RETRANS
2. RETRANS team establishes radio link with BNs from hilltop high ground using vehicle whip antennas with orders to displace every 10 minutes
3. Enemy direction finding teams locate RETRANS team and pass information to BTG TOC
4. BTG TOC passes location from direction finding teams to artillery battalion
5. Artillery battalion fires on last known location of RETRANS instead of BCT TOC which is able to remain stationary

Electromagnetic Protection TTPs (Emissions Control)

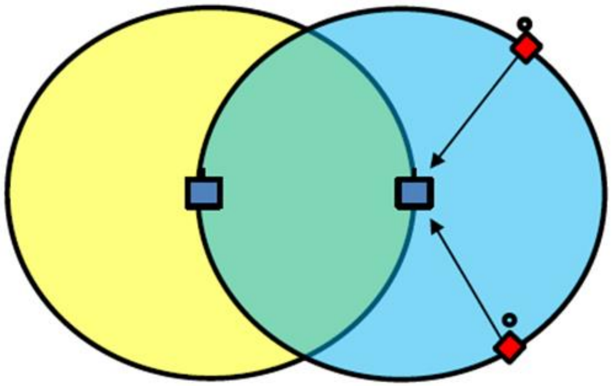
UNCLASSIFIED

Traditional TOC **VS** *Electromagnetically Protected TOC*



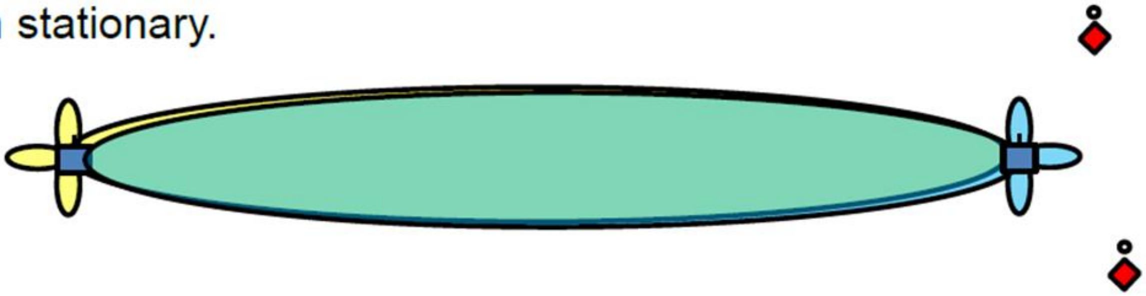
- What are the Pros vs Cons for each of these?
- Which one do units mostly use? Why?
- How would you present the importance of this or justify to your organization that one may be better for a given environment?

Electromagnetic Protection TTPs (Directional Antennas)



Using Omni-Directional Antennas emit a circular pattern as seen from above and you get 1 radius of communications distance but also a lot of energy emitted in areas you don't need. Use Omni-Directional (whip) antennas when mobile.

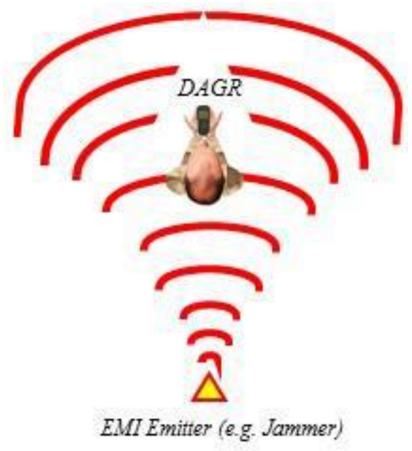
Using Directional Antennas you get more distance and most of your energy used along the communications channel. Use when stationary.



*Note, back and side lobes are not to scale and will vary depending on antenna design.

Personal Electromagnetic Protection TTPs

<u>Body Mass Shielding</u>



- Use your body to block the interference by placing your body between the offending signal transmitter (e.g. radar, jammer) and the receiver (e.g. DAGR)
- Hold receiver very close to your body with screen pointed away
- Rotate body slowly (~90 degrees) until receivers no longer affected by the EMI –allow sufficient time (~2 minutes) for receiver to acquire or lose signal from each position tested
- When signal is restored likely EMI source is behind you

<u>Terrain Masking</u>

- For mounted or dismounted operations, use natural terrain features (hills, buildings, rock formations) or create a barrier to block the offending signal
    - For best results, place receiver near the terrain feature
    - Interference will reappear the further away from the feature you move
- Smaller receivers not permanently affixed to a vehicle can be placed in a hole or inside a vehicle's hatch (simulates a hole) to block EMI and allow the GPS signal to be received
    - Hole should be 6-8 inches deep and allow sufficient view to the sky… do not stand over the hole – it blocks the signal
    - Allow time for the receiver to re-acquire the signal

Defeating Jamming

Electromagnetic Protection Helpful Tips

- Really great RETRANS sites are artillery magnets
  - Are you biting off more terrain that you can chew if COMMS go down?
  - Can SMs use the PACE place (i.e. rehearsals)?
- Do you need constant “On” communications, or should you use windows of clear comms during key times?
- Right tool for the right task/job WRT antennas. Do you need an amplifier (louder Tx) or a better antenna (better Tx and Rx)?
- ES assets properly located on the battlefield (i.e. not next to the transmitters or RETRANS station for the unit.
- EA for a purpose. Don’t Jam to Jam. Synchronize EA with maneuver/fires triggers for max effects.
- Jammers attack receivers. Units getting jammed can still Tx but may not be able to Rx. Distance will diminish jamming effects.

Electromagnetic Interference (EMI)

EMI prevents successful EM transmissions (Remember Signal to Noise Ratio)

- Units must recognize and mitigate EMI to create the conditions required to use the EMS to communicate
- Any electromagnetic disturbance Induced intentionally or unintentionally, that interrupts, obstructs, or otherwise degrades or limits the effective performance of Electromagnetics and electrical equipment.
- Can be induced intentionally, as in some forms of Electromagnetic warfare, or unintentionally, as a result of spurious emissions and responses, intermodulation products
- Once EMI is identified, a Joint Spectrum Interference Report (JSIR) is initiated and forwarded through spectrum manager channels

Defeating Jamming (1 of 2)

Defeating Jamming:

- The adversary uses diverse forms of jamming
- Radio operators require awareness to the possibility of jamming.

Threat EA usually involves jamming followed by a brief listening period

Operator activity during the listening period indicates how effective the jamming to alleviate the threat has been

Defeating Jamming (2 of 2)

Improve the Signal to Jamming Ratio

- The signal-to-jamming ratio is the relative strength of the desired signal to the jamming signal at the receiver

To improve the signal-to-jamming ratio operators should:

- Increase the transmitter power output
- Adjust, change or relocate the antenna
- Establish a retransmission station
- Use an alternate route for communications
- Change frequencies

Reporting Jamming

Reporting Jamming

- Units report suspected threat jamming and any unidentified or unintentional EMI that disrupts the ability of U.S. forces to communicate

- Units report any suspected jamming or EMI

  - Even if the radio operator can overcome the effects of the jamming or EMI

- All interference is reported on a Join Spectrum Interference Report (JSIR)

- **Operators must be trained to identify jamming if they are going to report it!**

Common Jamming Signals (1 of 2)

Random Noise:

- It is indiscriminate in amplitude and frequency
- Sounds like normal background noise

Stepped Tones:

- Tones transmitted in increasing and decreasing pitch
- Resemble the sound of bagpipes

Spark:

- One of the most effective jamming signals
- Spark uses short intensity and high intensity; they repeat at a rapid rate.

Common Jamming Signals (2 of 2)

Gulls:

- Generated by a quick rise and slow fall of a variable radio frequency
- Similar to the cry of a sea gull

Random Pulse:

- Pulses of varying amplitude and duration are generated and transmitted.
- Disrupt teletypewriter, radar, and all types of data transmission systems.

Wobbler:

- Single frequency, modulated by a low and slowly varying tone.
- Howling sound that causes a nuisance effect on voice radio communications.

Preamble Jamming:

- Broadcasted tone over the operating frequency of secure radio nets
- Resembles the synchronization preamble of the speech security equipment

EP Planning

Planning to Succeed

- While there are a wide variety of EP techniques, the most successful units train on and incorporate EP into their plans long before conducting an operations

- EP is as much (if not more) about the **culture and discipline of a unit** as it is the technical capabilities and vulnerabilities of their equipment

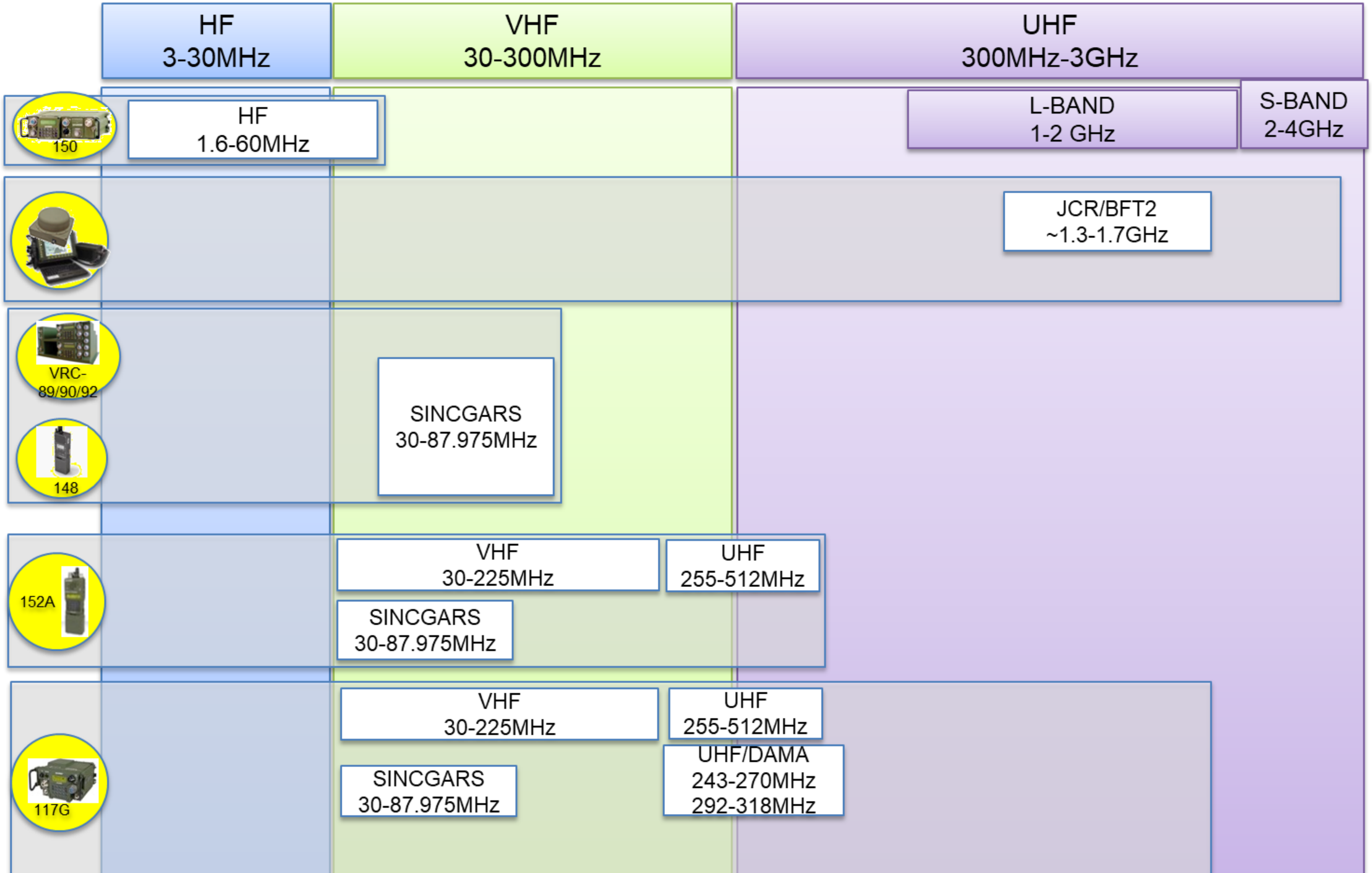
- **EP is a team effort**. The CEMA Cell cannot control EP single handedly, they must get the total buy in of the unit.

- **EP is the single most important task of BDE and below CEMA and the hardest**

Planning to Succeed

- Review your communications network with your S-6
- Review and understand threat systems and capabilities with the S-2
- Review reporting and battle rhythm procedures with your S-2
- Conduct an EM vulnerabilities assessment with your S-2 and S-6
- Make an Updated PACE plan based off of a vulnerabilities assessment
- Incorporate Jamming in all collective training events
- Regularly review and training interference reporting procedures
- Conduct Emission simulations with virtual planning Tools prior to moving into an area
- Identify major terrain features for masking or observations

EXAMPLE
Friendly EMS Assessment



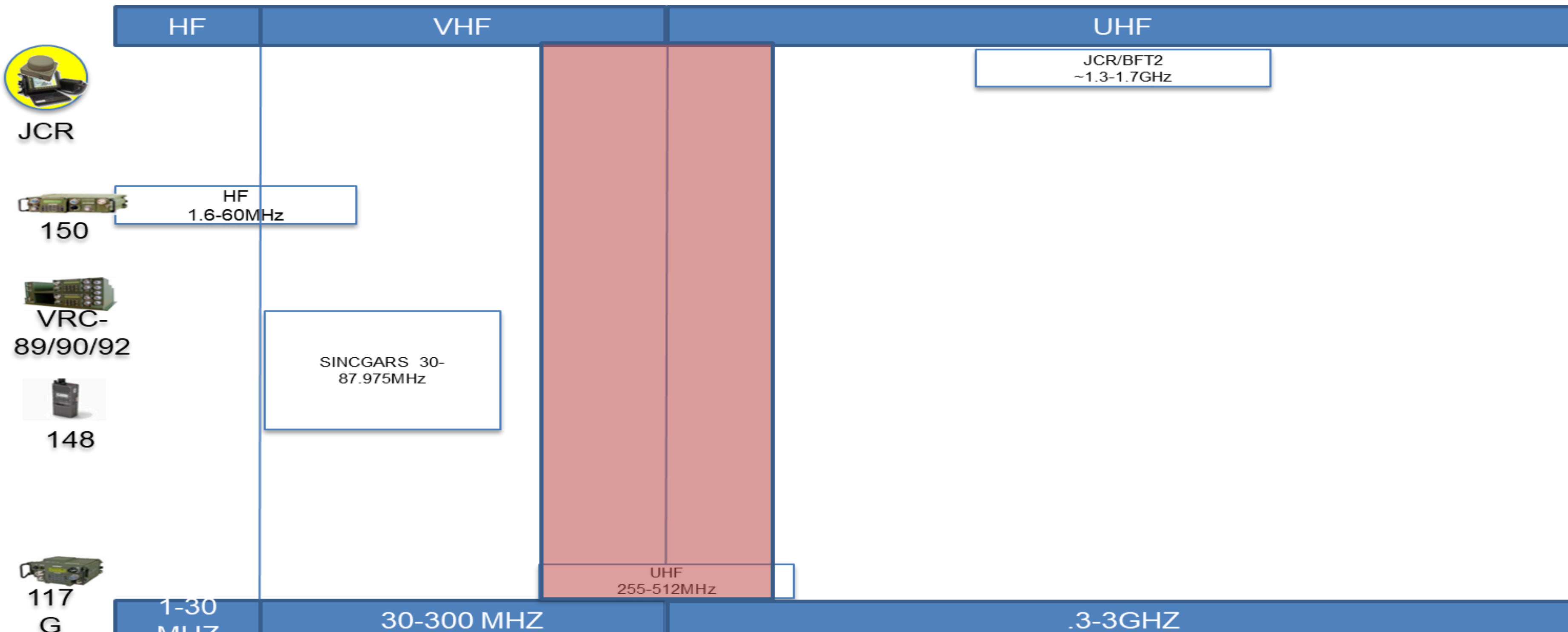
EXAMPLE
Enemy Threat Assessment

Aviaconversia EA system

A man/Vehicle portable, Russian made UHF jammer.

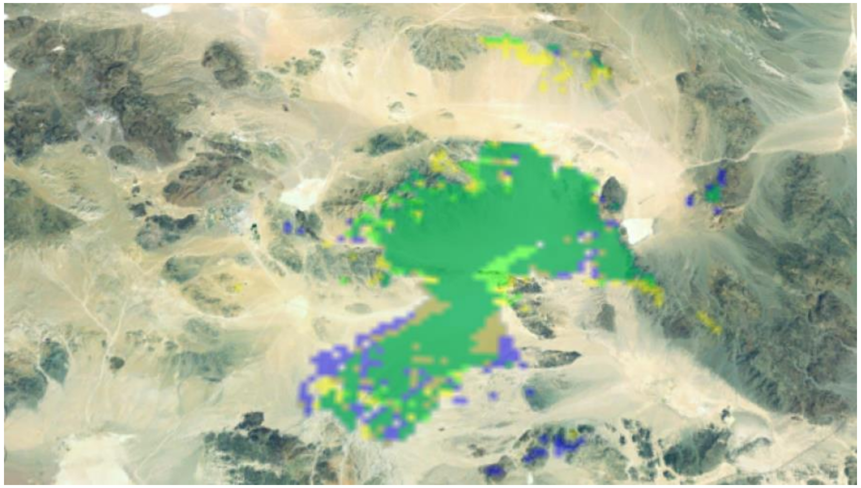
Characteristics:
-Jamming Freqs:230MHz 433MHz
-Power:4W
-Range:150-200KM

Likely Number: 3 (1 Per recon Company)



Modeling and Simulations

- Can be used to plan and evaluate EMS actions against real terrain data
- Helps identify key terrain (for EMS signal) as well as potential obstacles
- Great for showing areas of coverage and masking for EMS signal
- Multiple programs available
  - Electronic Warfare and Planning Management Tool (EWPMT)
  - Systems Planning, Engineering and Evaluation Device (SPEED)
  - GPS Interference and Navigation Tool (GIANT)



Summary

- Define Electromagnetic Protection.
- Describe the fundamentals of (EP).
- Describe EP tasks
- Describe EP TTPS
- Describe Electromagnetic Interference Reporting
- Describes EP planning considerations

Questions?

References

- FM 3-12 Army Cyberspace and Electromagnetic Warfare Operations. Apr 17.
- ATP 3-12.3 Electromagnetic Warfare Techniques.
- JP 3-13.1 Electromagnetic Warfare

Electromagnetic Support

112-BOLCEW03

Enabling Learning Outcome

The Student will define divisions of Electromagnetic Warfare doctrine, in a classroom environment, given a network computer workstation, training aids, and appropriate references by demonstrating understanding of the divisions of Electromagnetic Warfare Doctrine; Electromagnetic attack, Electromagnetic protect, and Electromagnetic support in order to support the application of EW effects critical to the units mission in accordance with JP 3-13.1, ADRP 6-0, FM 3-12, FM 6-0, ATP 3-36, while achieving a minimum of 70% on all Practical Exercises (PE) and end of module exam.

Learning Step Activities

- Define Electromagnetic Support (ES).
- Describe the fundamentals of ES.
- Compare and Contrast ES and SIGNT
- Define ES Tasks
- Describe ES TTPS

Principles of Electromagnetic Warfare Support (ES)

Electromagnetic Warfare Support

Electromagnetic Warfare Support (ES) involves actions tasked by, or under direct control of an operational commander to search for intercept, identify, and locate or localize sources of intentional and unintentional radiated electromagnetic energy

Collected information is used for the purpose of immediate:

- Corroborate other sources of information or intelligence
- Conduct of EA operations: Cueing of EA systems to targets
- Self-protection and EP measures
- Targeting
- Support IRCs: IIA, MILDEC, MISO, OPSEC, etc

Electromagnetic Warfare Support

Commanders require the detection and location of emitters to:

- Aid in targeting
- Provide information to the unit for potential inclusion in intelligence gathering by SIGINT professionals.

"ESM/ES, on the other hand, collects enemy signals (either communications or non-communications) with the object of immediately doing something about the signals or the weapons associated with those signals. The received signal might be jammed or its information handed off to a lethal response capability. The received signals can also be used for situation awareness; that is, identifying the types and location of the enemy's forces, weapons, or Electromagnetic capability. ESM/ES typically gathers lots of signal data to support less extensive processing with a high throughput rate. ESM/ES typically determines only which of the known emitter types is present and where they are located."

David Adamy (author EW 101/102)

Electromagnetic Warfare Support Actions

Electromagnetic Reconnaissance

- Detection, location, identification, and evaluation of foreign electromagnetic radiations

Electromagnetic Intelligence

- Technical and geo-location intelligence derived from foreign non-communications electromagnetic radiations

Electromagnetics Security

- The protection resulting from all measures designed to deny unauthorized persons information of value that might be derived from their interception and study of non-communications electromagnetic radiations.

ES and SIGNT

Though EW and SIGINT are similar, there are important distinctions between them. Legal considerations distinguish EW and SIGINT activities, and the authorization for each to support operations, that if not observed, can complicate and delay the execution of electronic warfare effects and SIGINT operations. (ATP3-12.3)

- ES falls under title 10
- SIGNT fall under title 50
- ES uses information to directly support operations ie(targeting or PIRs)
- SIGNT uses information to build intelligence products
- ES systems can pass information to SIGNT systems
- Some "sterilized" information can be passed to ES systems from SIGNT
- ES and SIGNT are often co-located and directly support one another
- The **task and purpose** of the mission determines what assets should be used

Integrating EW and SIGINT is a force multiplier for unified land operations. EW and SIGINT have similar capabilities that are mutually beneficial. Integrated EW and SIGINT assets present an efficient, holistic approach that reduces duplication of effort, enables additional information collection, and provides flexibility in the employment of EW and SIGINT resources. (ATP3-12.3)

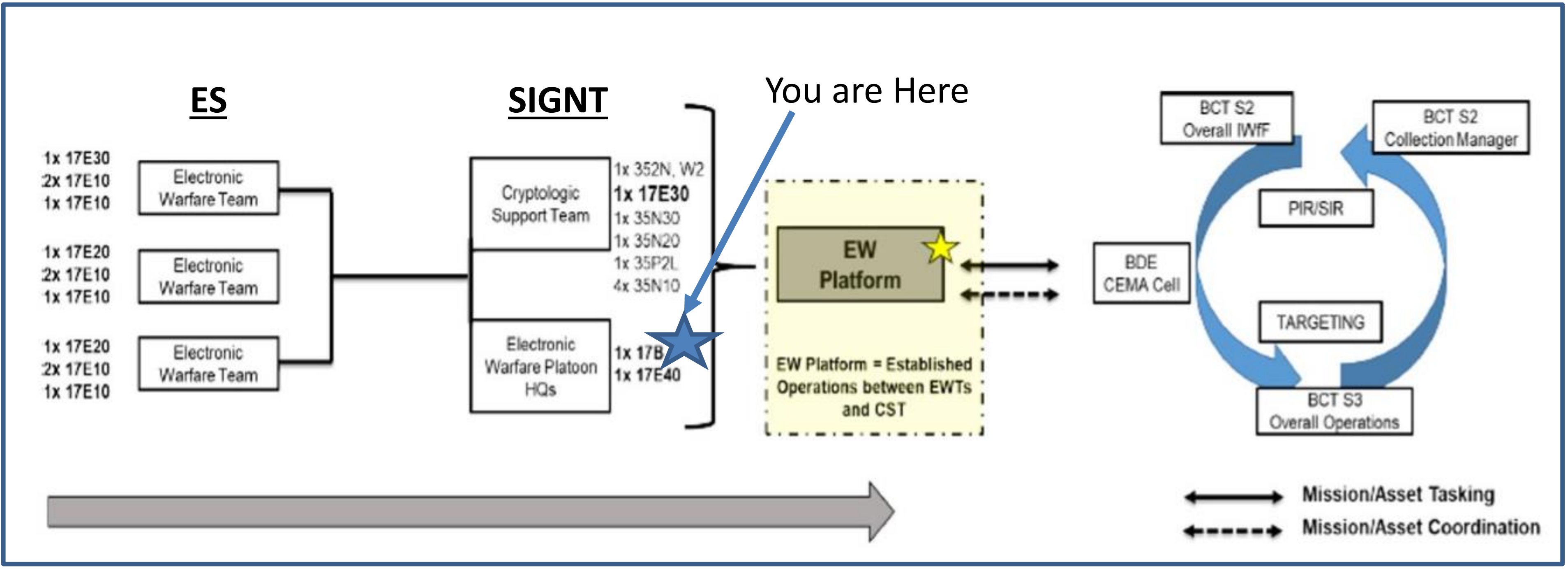
ES and SIGINT

Q:Why care about the difference between ES and SIGNT?

A: Because the average EW Platoon leader resides in the Military intelligence company!

A: Because not understanding the difference can get you in trouble!

Structure and Flow of a BDE EW PLT



Electromagnetic Warfare Support Considerations

The commander's task and purpose determines if the mission contains Electromagnetic Warfare Support (ES)

Commanders allocate assets to conduct ES for immediate:

- Threat recognition, targeting, future operations Planning
- Threat geolocation
- Threat avoidance
- Answering information requirements

ES Techniques

ES Preparation

Electromagnetic Environment Survey

- Like weather reports for aircraft pilots, the EME survey informs the CEWO about the activities and conditions of the EME, enabling the CEWO to choose optimal COAs for EW. 5-9.

- EME surveys begin with the **enemy Electronic Order of Battle (EOB)**. The enemy EOB provides the CEWO with an initial overview of threat EMS capabilities derived from IPB. The enemy EOB assists the CEWO in making EW plans that exploit adversary vulnerabilities while preserving friendly capabilities. The enemy EOB is the baseline for the EME survey

Site Selection

- Site selection is the process of determining the best location to conduct

ES Preparation

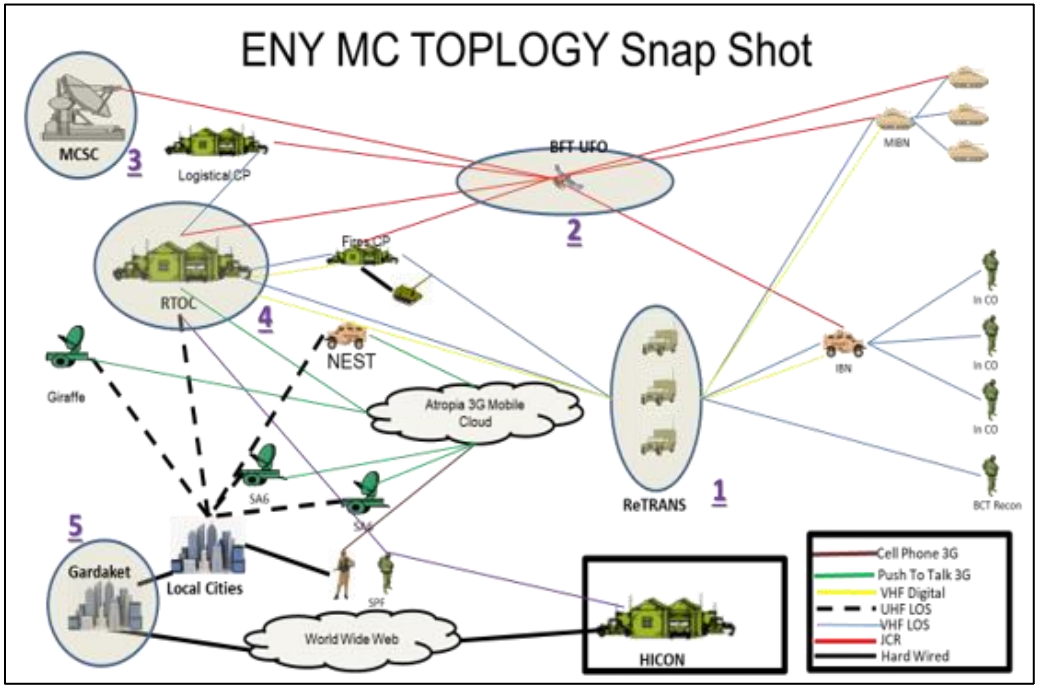
Site Selection

- Site selection is the process of determining the best location to conduct ES operations.
- Site selection is determined off of many factors to include:
  - Terrain features and Elevation
  - Enemy avenues of approach
  - Coverage and reception of signals of interest (Based off the EOB)
  - Communication coverage and support
  - Concealment of site
  - Logistical support (depending on duration)
- Basic tools required for site selection include
  - Map
  - EOB
  - RF simulation software or prebuilt coverage maps
  - Enemy and friendly SITEMPS

ES Preparation

Electronic Order of Battle

- A comprehensive document that lays out enemy electronic systems with associated platforms and frequencies
- Created obtained by the **SIGNT TECH** within the MICO
- Can be a simple list of frequencies and systems or a logical layout of a RF network
- **One of the most single important documents for an EW PLT LDR**



Direction Finding

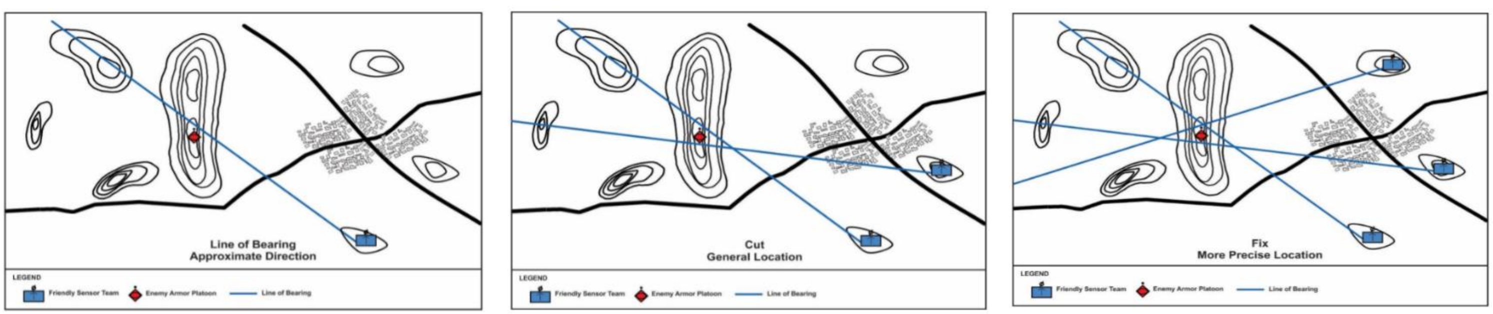
- **Direction Finding (DF)** is the act of utilizing RF equipment to locate signals of interest.

- DF provides Line of Bearings (LOBs) which can be used to make Cuts, and Fixes which can locate transmitters

  - LOB - Single DF sensor providing the approximate azimuth to the transmitter.

  - Cut - Two DF sensors providing the general location of a transmitter by determining where two LOBs intersect.

  - Fix - Three or more DF sensors providing a location using a triangulation method. direction.



Direction Finding

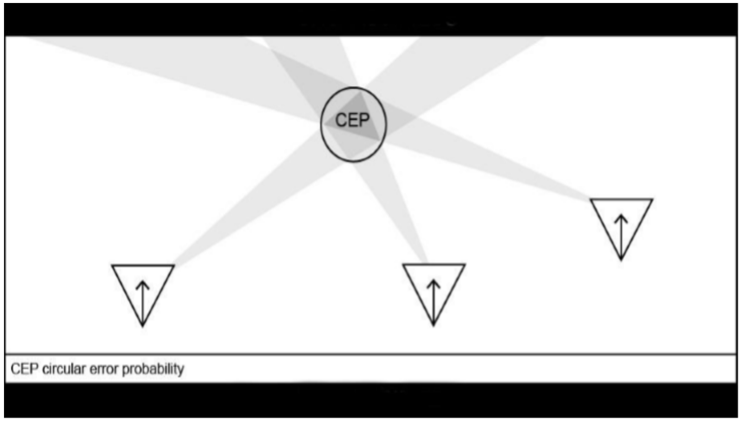
Line of Bearings (LOB) to Target

The reception of a LOB on a target signal by a DF receiver does not always determine the exact azimuth to a transmitter. This is due to:

- Weather and terrain
- Equipment accuracy
- Transmitter distance

When LOBs are plotted, there is a triangular area of overlap where the LOBs intersect to form a fix.

- Circular error probability is the precise location of the transmitter cannot be certain.



CEP circular error probability

Direction Finding

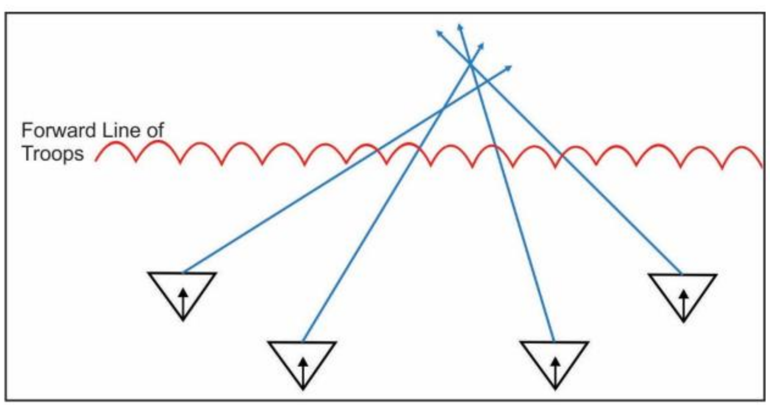
- **Path Error**: Deviations between the transmitter and DF system are
- Important sources of path error include:
  - Scatter is a small portion of the radio wave entering the ionosphere is scattered to the Earth at random angles
  - Refraction occurs when waves bend or refract from their normal path as they pass from one medium to another
  - Reflection occurs when radio waves strike and reflect off an artificial or natural surface
  - Reradiating occurs when a wave strikes a metallic object that resonates at the wave's frequency

Direction Finding

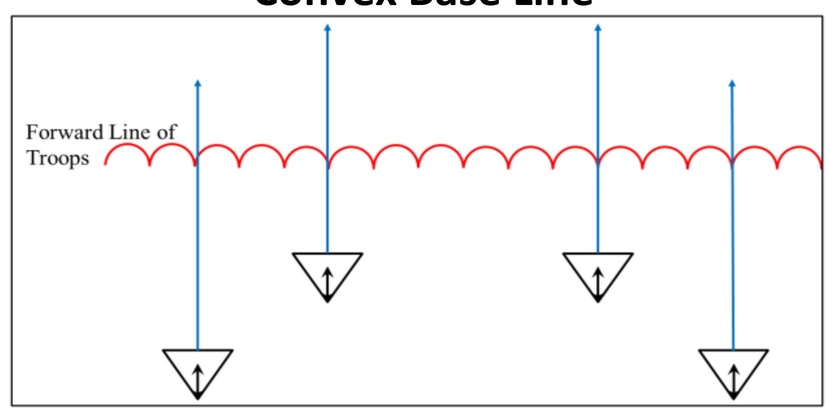
Understanding your Equipment:

- ES systems utilize simple DF techniques ie(just LOBs) IOT locate sources of signals
- Gain and Sensitivity can differ between systems and antennas
- Most ES systems utilize advanced DF methods such as Time Difference of arrival (TDOA) or Correlated Informatory (CI) to produce location data
- **BLUF:** How your equipment works should significantly factor in to how you deploy it!

Concave Base Line



Convex Base Line



Summary

- Define Electromagnetic Support (ES).
- Describe the fundamentals of ES.
- Compare and Contrast ES and SIGNT
- Define ES Tasks
- Describe ES TTPS

Questions?

References

- FM 3-12 Army Cyberspace and Electromagnetic Warfare Operations. Apr 17.
- ATP 3-12.3 Electromagnetic Warfare Techniques.
- JP 3-13.1 Electromagnetic Warfare